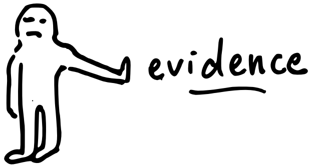
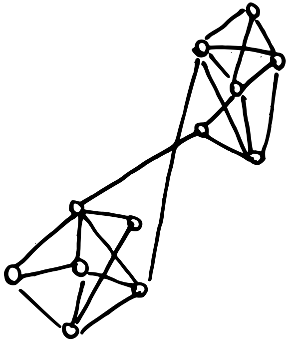
# Confirmation bias and polarization




Clémence Bergerot
Romanczuk Lab (Humboldt University Berlin)
& Barfuss Lab (University of Bonn)

Talk at Forward College
22.10.2025



HUMBOLDT-UNIVERSITÄT ZU BERLIN

# Polarization in contemporary democracies

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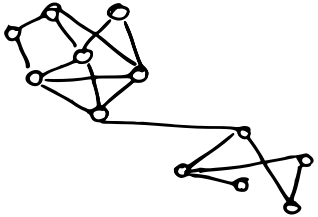
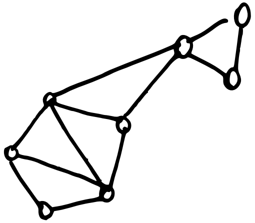
**Opinion polarization:** an increase in antagonistic and extreme opinions across members of a population

# Polarization in contemporary democracies

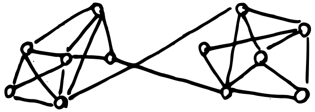

US Capitol invasion, Jan 6th, 2021



Brazilian Congress Attack, Jan 8th, 2023




Brexit, 2020



*Images: New York Post, The Guardian, Business Insider*

# Polarization: why is it a problem?

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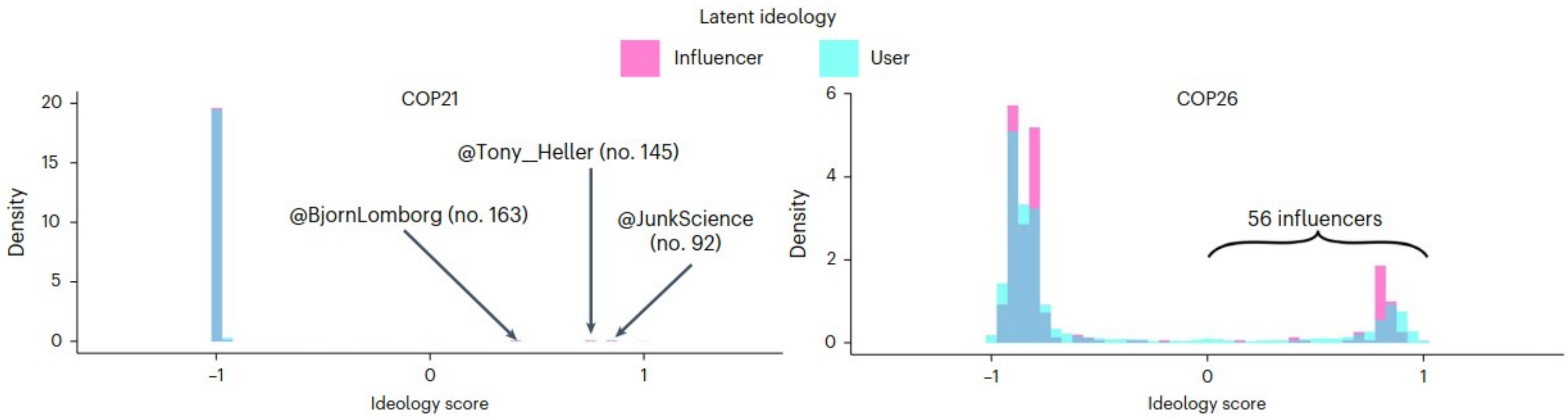


Increasing polarization about climate change on Twitter between COP21 and COP26.
From Falkenberg *et al.*, 2022.

# How can we explain polarization?

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**environment**

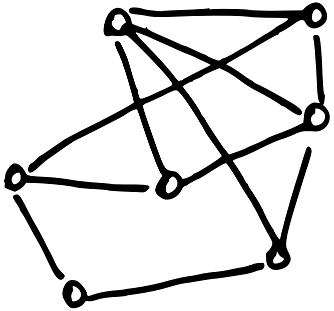
# How can we explain polarization?

environment


Information overload, recommendation algorithms...

# How can we explain polarization?

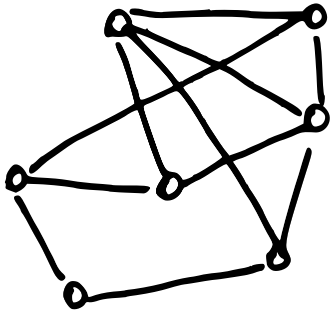
environment



social structure

# How can we explain polarization?

Echo chambers,
recommendation
algorithms…

**social structure**

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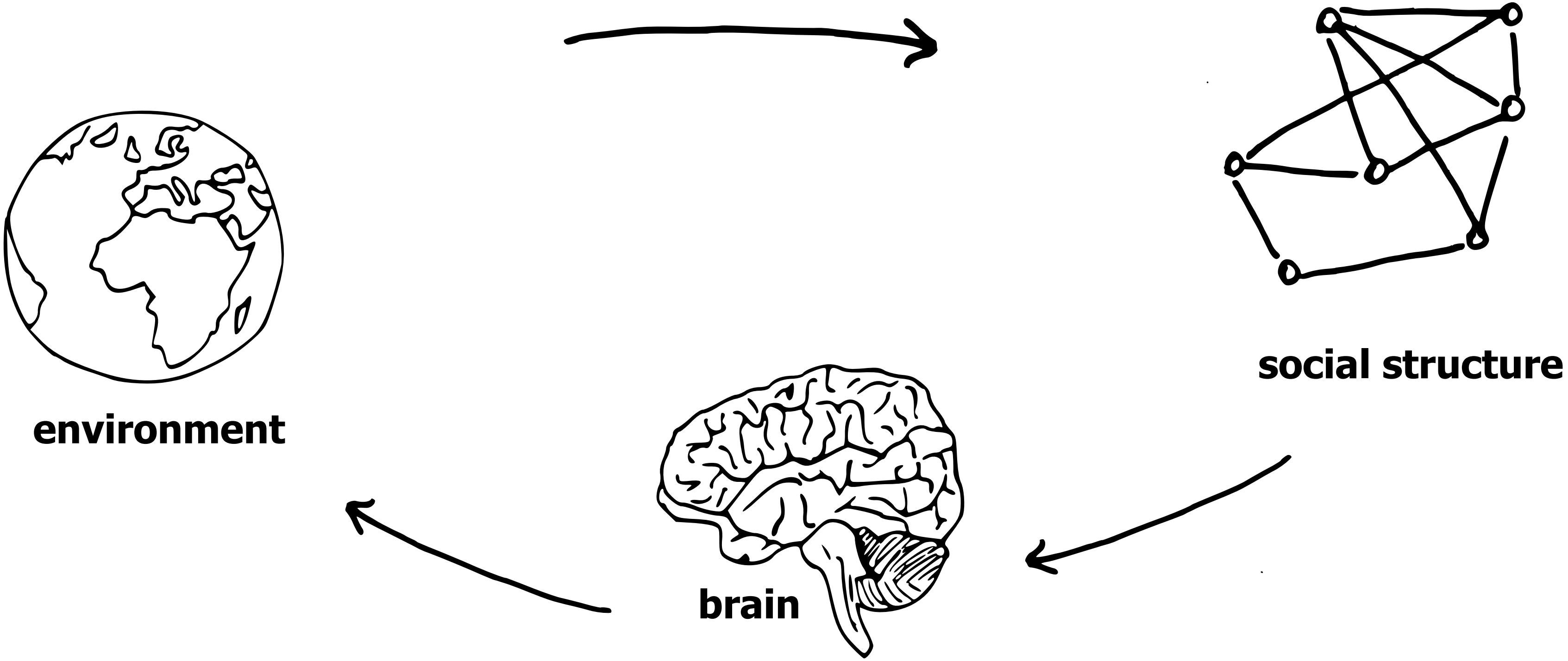

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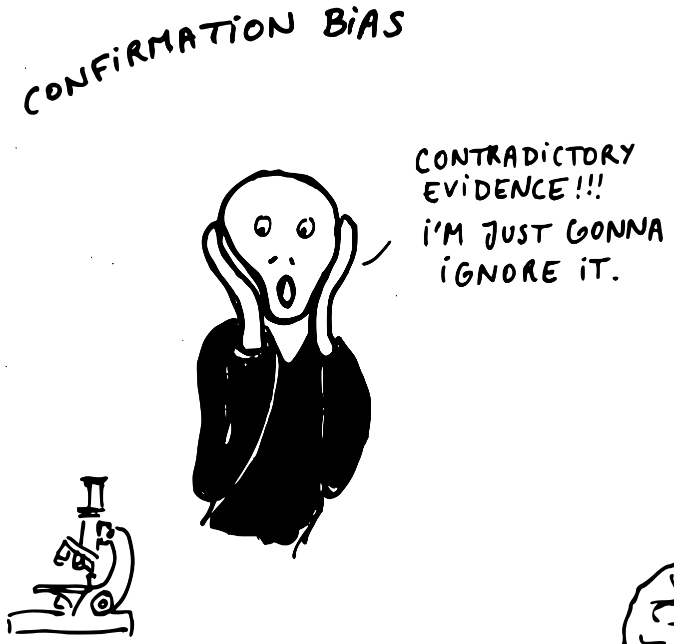
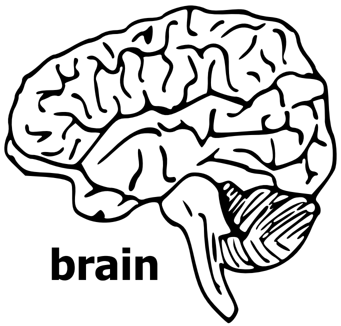



**Confirmation bias**: accept information that confirms one's prior beliefs, and dismiss information that contradicts them



# Research question

*How does the combination of confirmation bias, social interactions, and environment, give rise to polarization?*

# Computational neuroscience & collective behavior

**Collective Information Processing Group**

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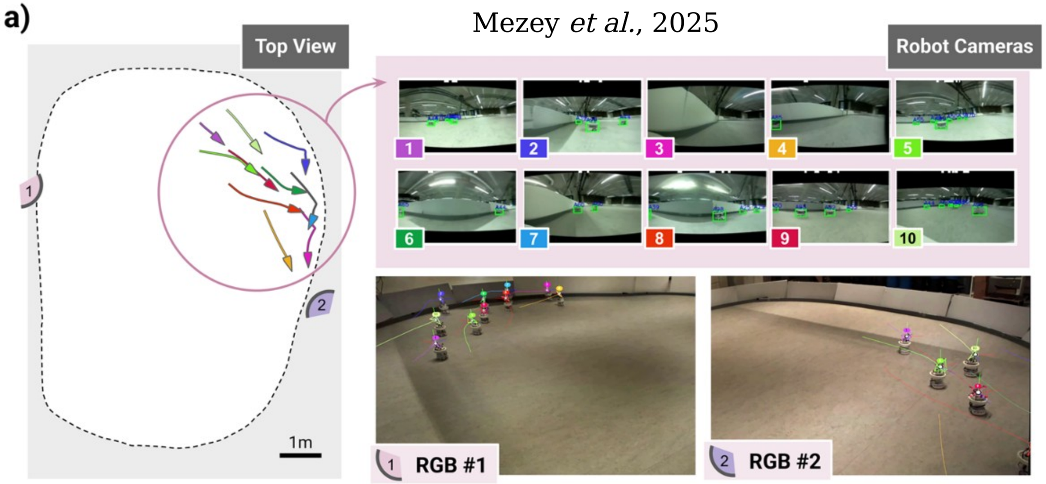
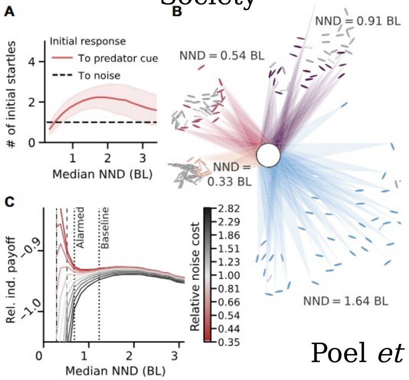



Mezey *et al.*, 2025

Pawel Romanczuk




Poel *et al.*, 2022

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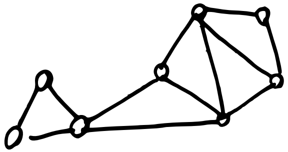

**Computational neuroscience**: developing models & algorithms to better understand cognition/brain processes

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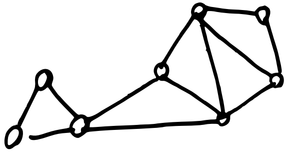

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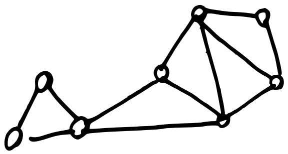


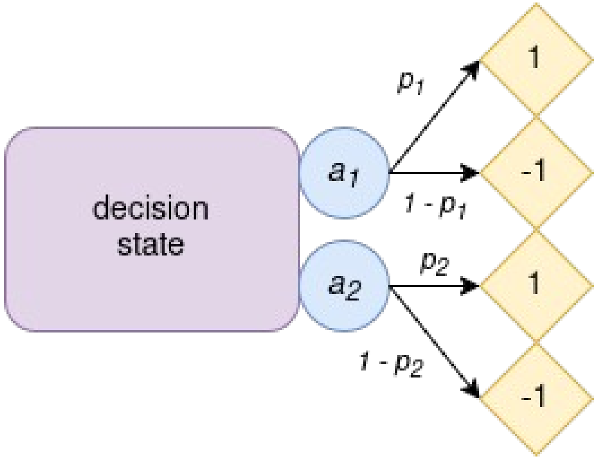
**Computational neuroscience**: developing models & algorithms to better understand cognition/brain processes

**Collective behavior**: bridging the gap between cognitive & collective levels

Toolbox: models, simulations

# Moderate confirmation bias enhances collective learning in groups of reinforcement-learning agents

# The two-armed bandit task



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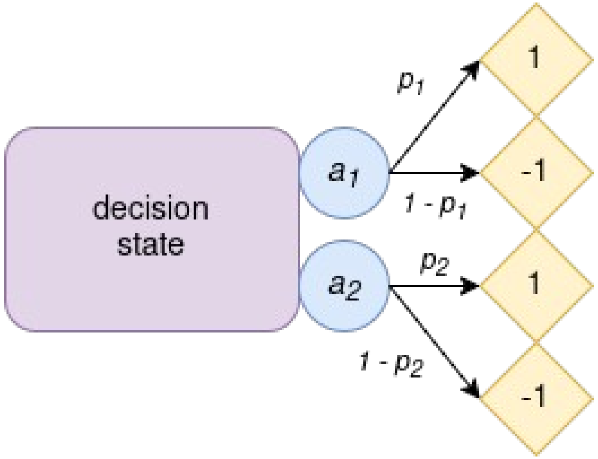
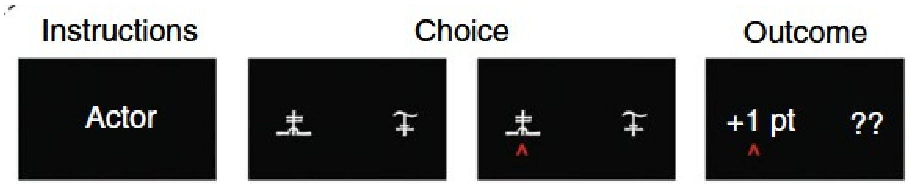
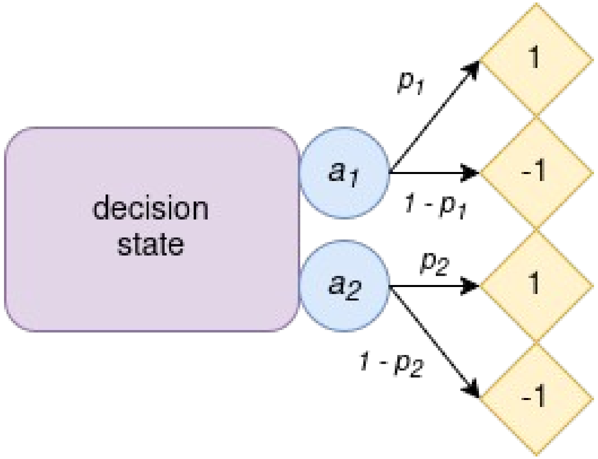
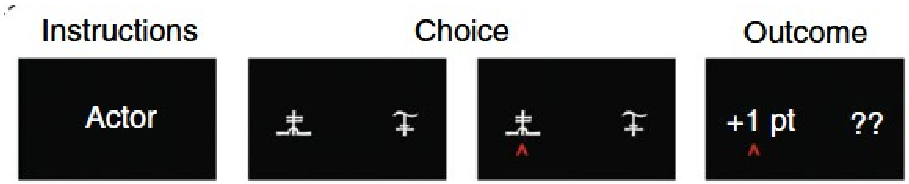



Chambon *et al., 2020*

# The two-armed bandit task

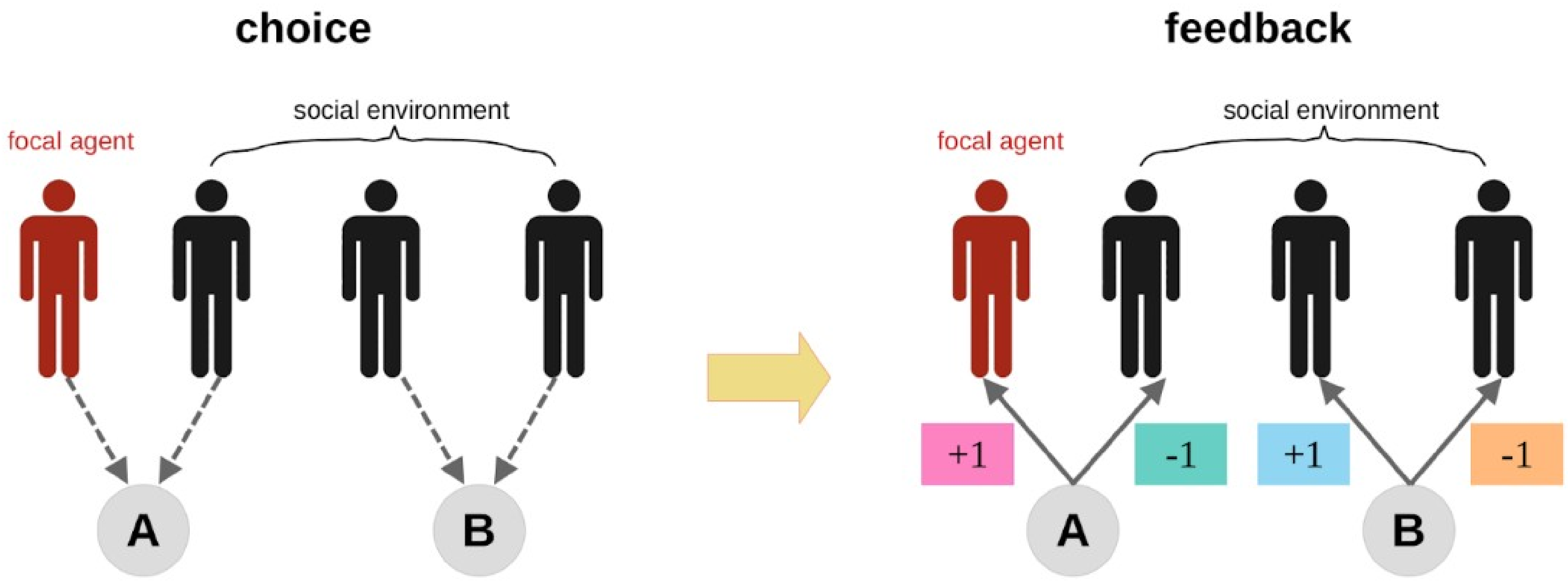



Chambon *et al., 2020*

| **Rich environment** | **Mixed environment** | **Poor environment** |
|---|---|---|
| $p_1 = 0.9$<br>$P_2 = 0.7$ | $p_1 = 0.6$<br>$P_2 = 0.4$ | $p_1 = 0.3$<br>$P_2 = 0.1$ |

# Confirmation bias in a collective setting




**Focal agent's Q-value update**

$$Q_{t+1}(A) = Q_t(A) + \frac{1}{4} \cdot [\alpha^{+}(\boxed{+1} - Q_t(A)) + \alpha^{-}(\boxed{-1} - Q_t(A))]$$

$$Q_{t+1}(B) = Q_t(B) + \frac{1}{4} \cdot [\alpha^{-}(\boxed{+1} - Q_t(B)) + \cdot\alpha^{+}(\boxed{-1} - Q_t(B))]$$

# Results

Confirmation bias impedes performance in small groups in rich environments

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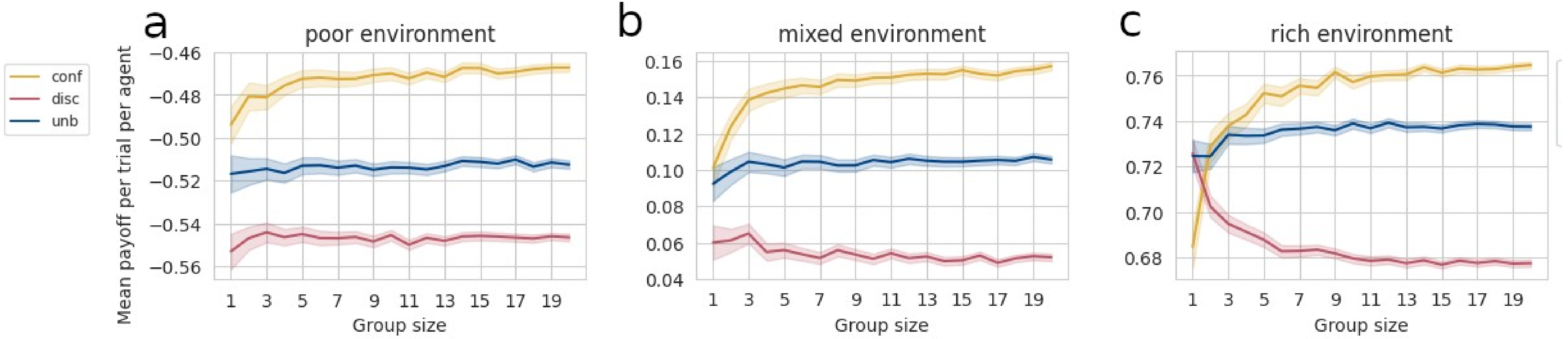

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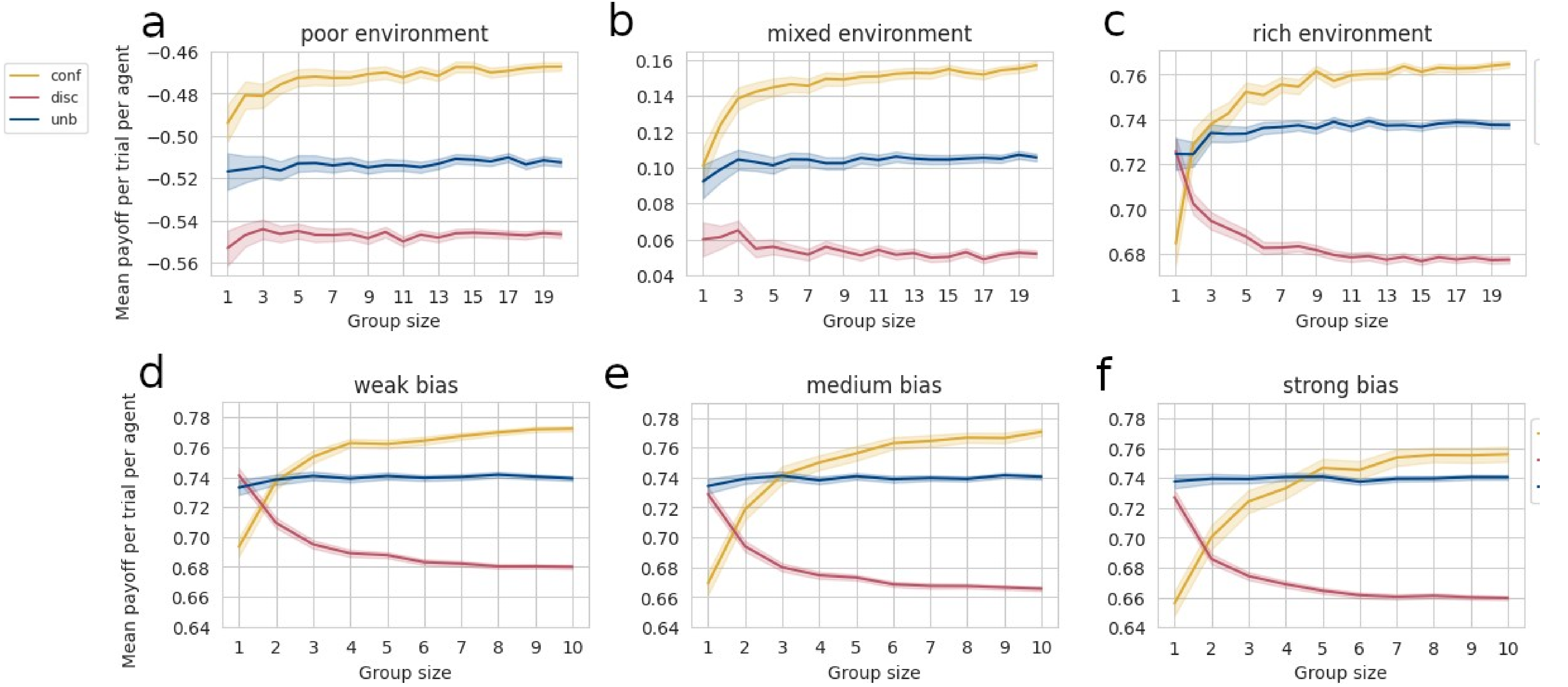

Confirmation bias causes the emergence of two different regimes of performance in small groups

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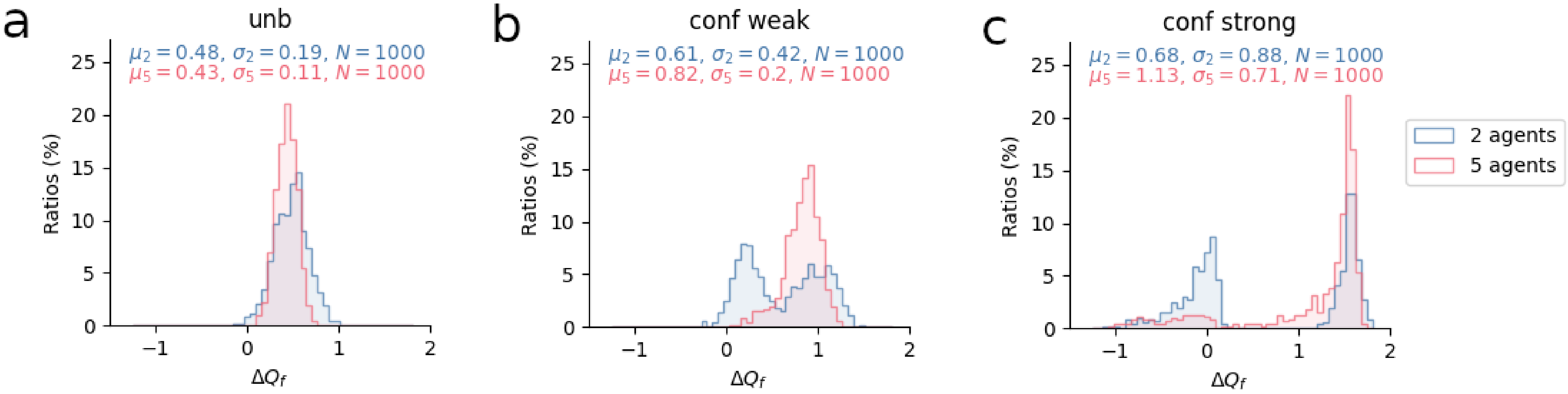

# In mixed and rich environments, moderate confirmation bias maximizes performance

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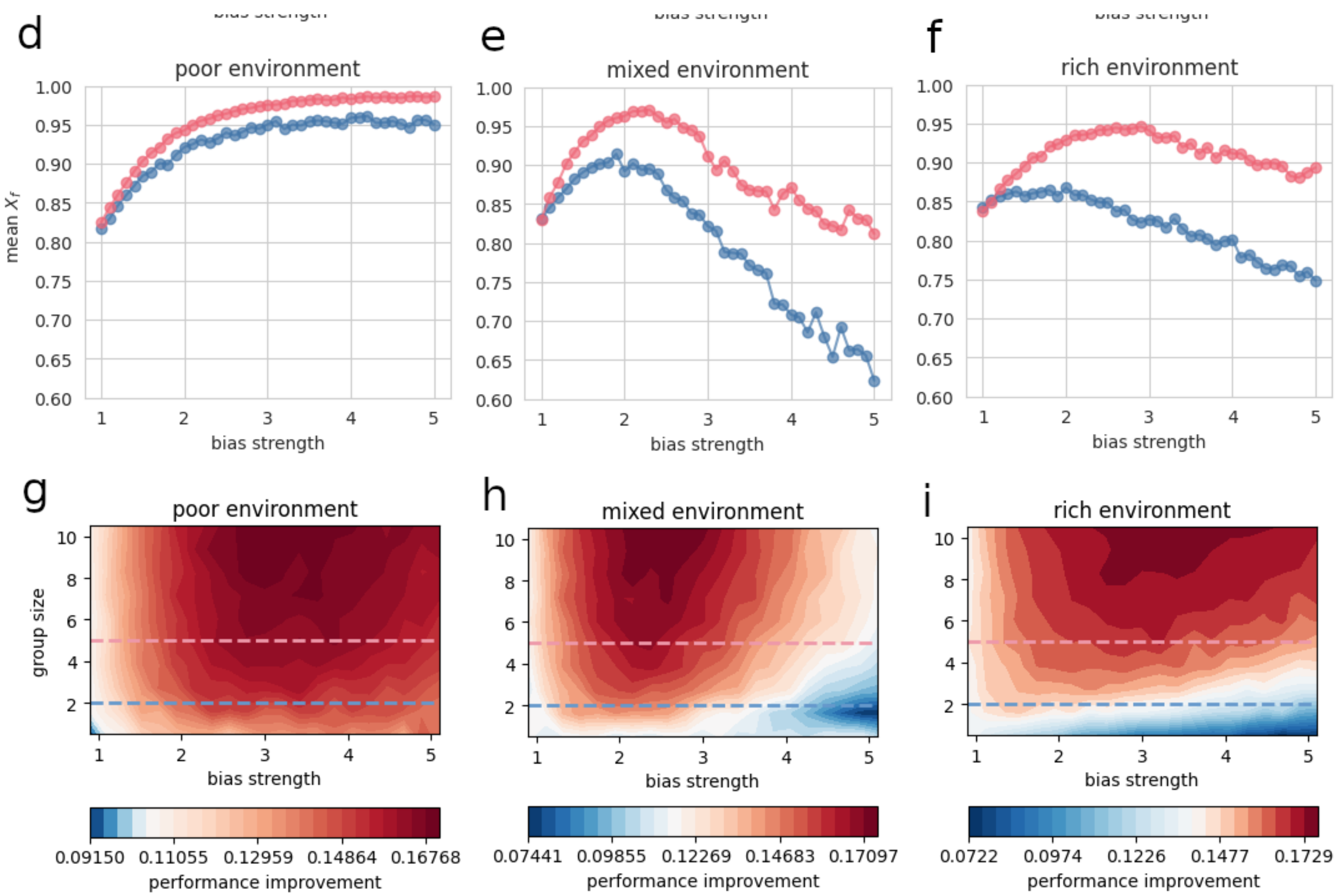

# Past a given bias strength, polarization emerges in mixed and rich environments

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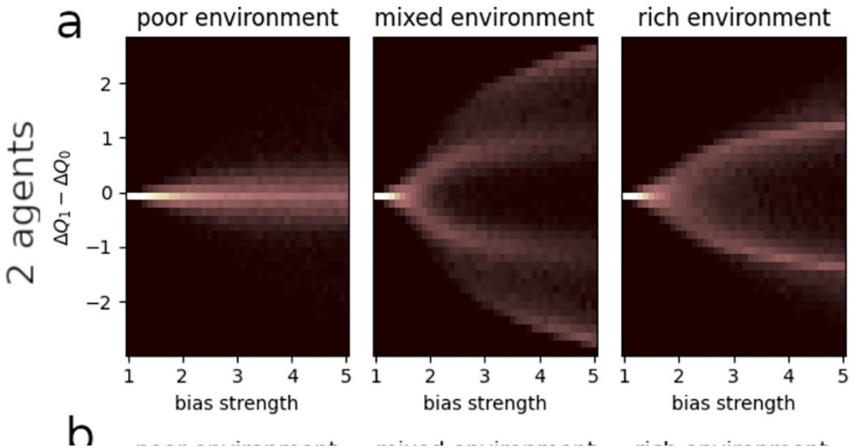

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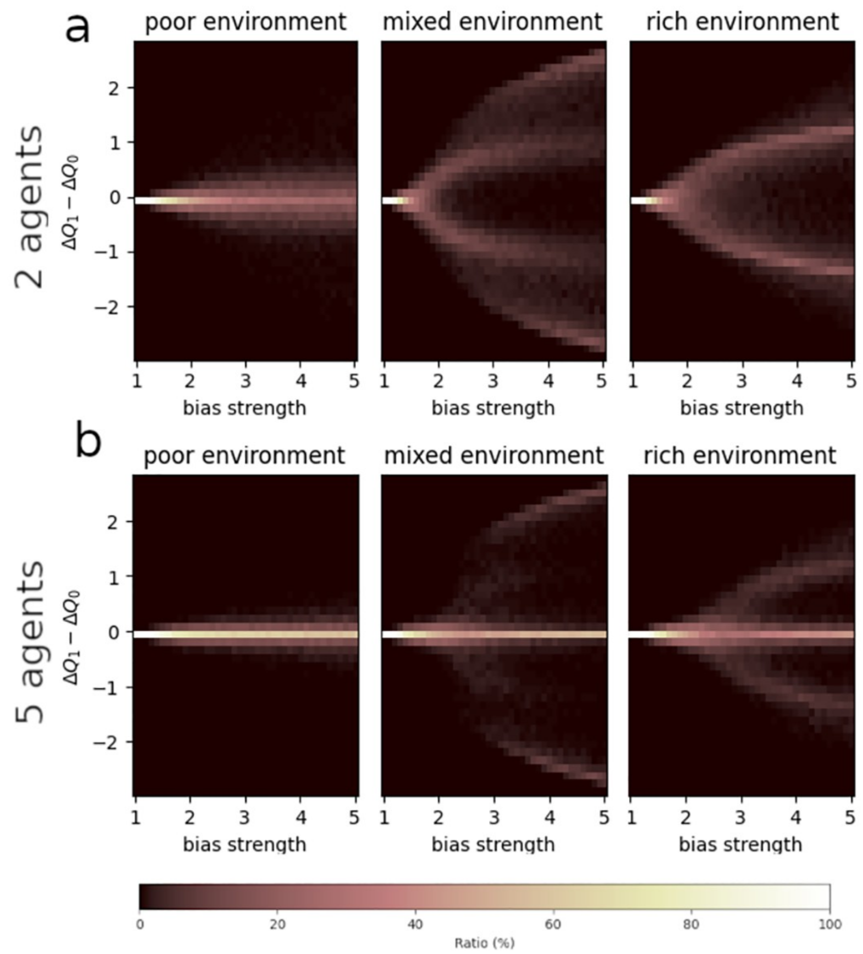

# Conclusion

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- past a given bias strength, polarization emerges and underlies a drop in performance
- optimal confirmation bias is moderate
- points towards adaptive value of confirmation bias in a social context

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- problematic effects could be due to strong confirmation bias, or to environment structure (e.g., filter bubbles)

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- problematic effects could be due to strong confirmation bias, or to environment structure (e.g., filter bubbles)

- What are the limitations of this model? What are potential next steps?

# Thank you for your attention!

EINSTEIN CENTER Neurosciences

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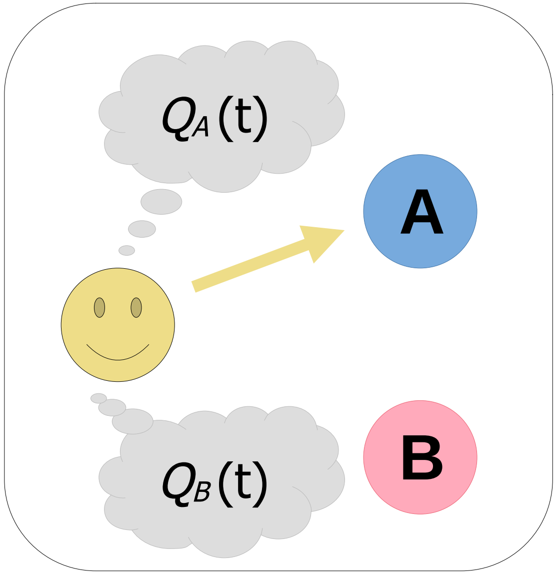
# Confirmation bias in reinforcement learning

Value estimates $Q_i(t)$



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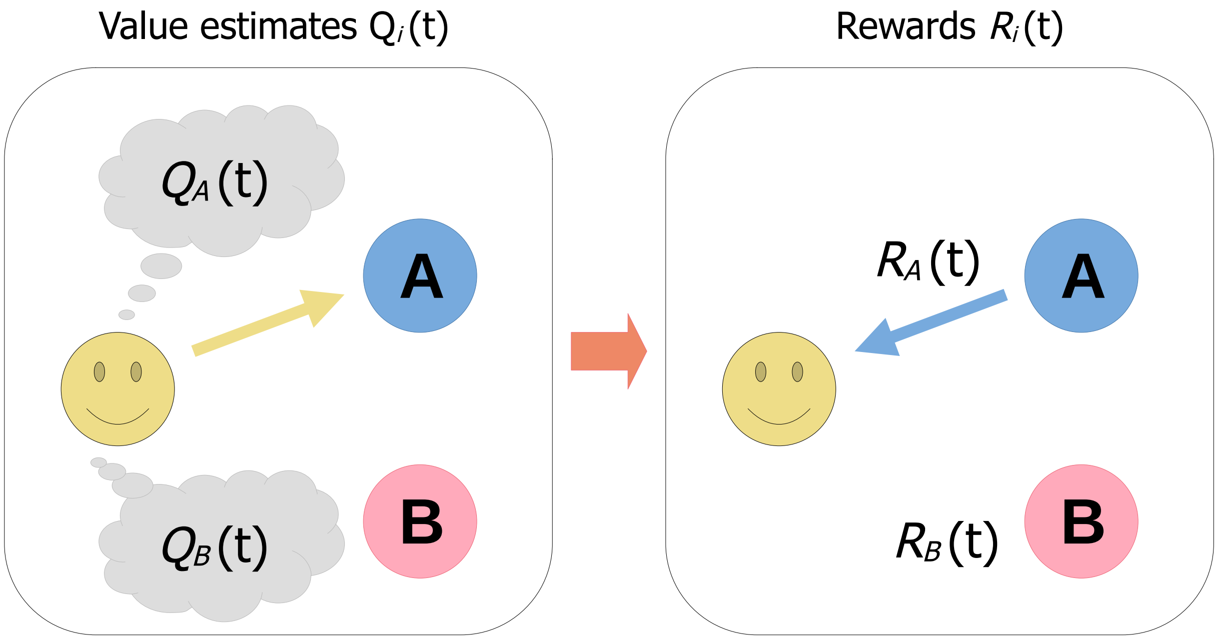

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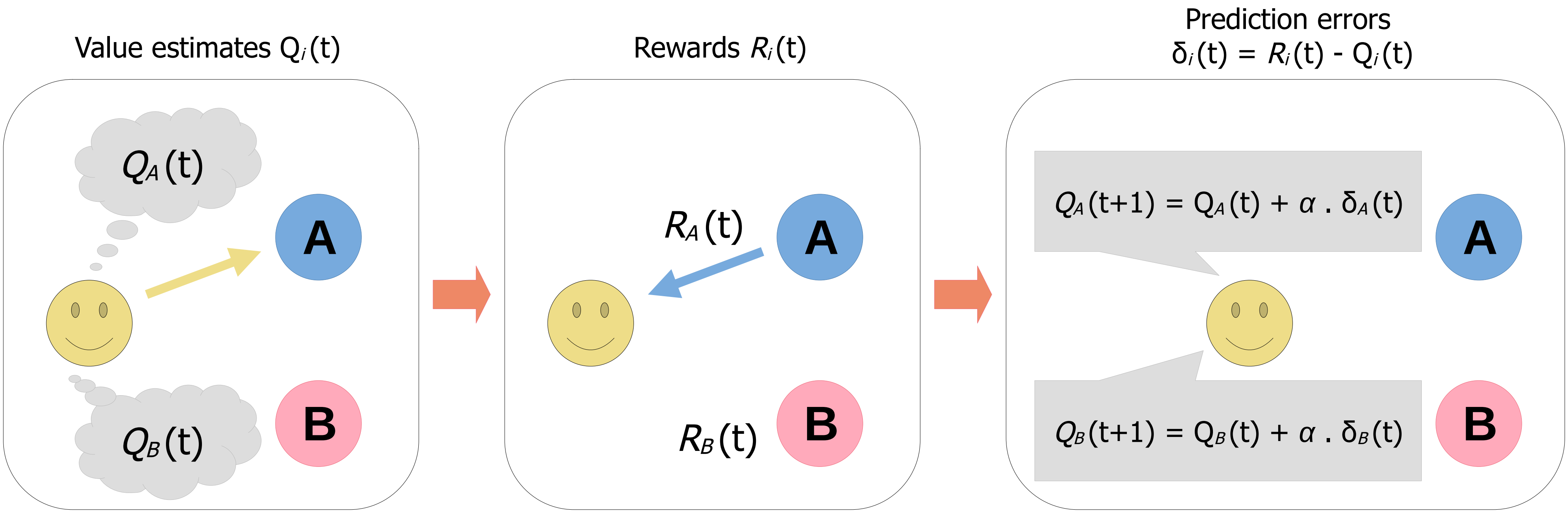

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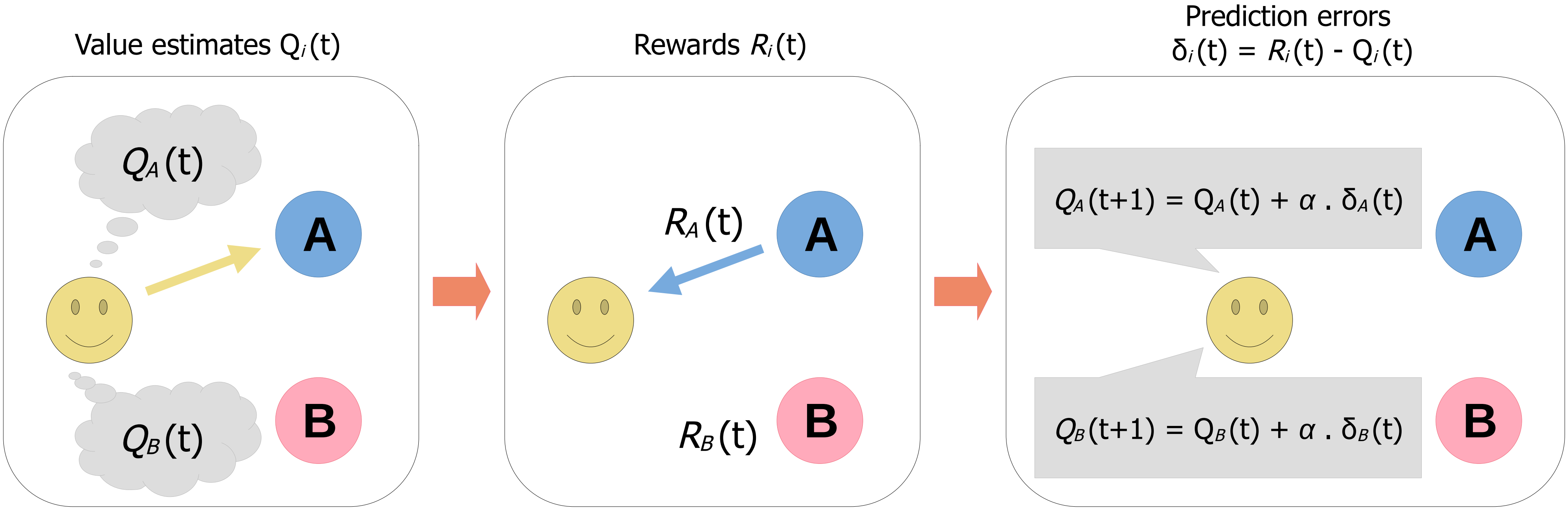


2 learning rates:

- $\alpha^+$ for confirmatory prediction errors
- $\alpha^-$ for disconfirmatory prediction errors
- $\alpha^+ > \alpha^-$

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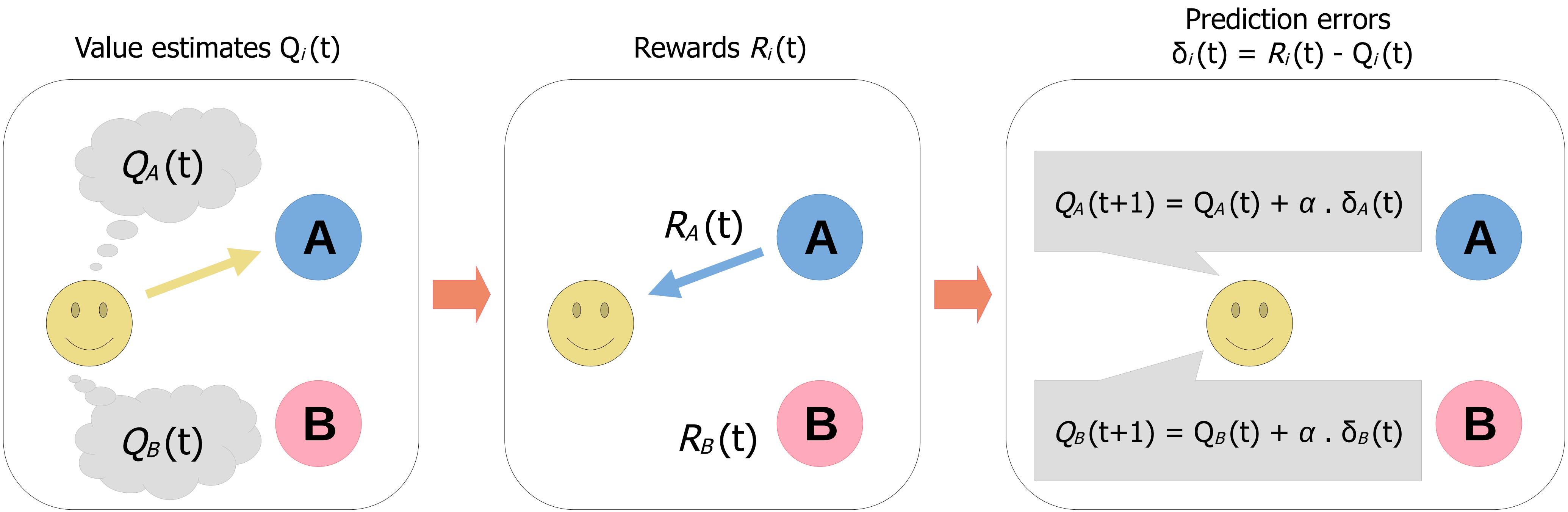


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confirmatory prediction errors:

- $\delta_i(t) \geq 0$ and *i* was chosen
- $\delta_i(t) < 0$ and *i* was unchosen

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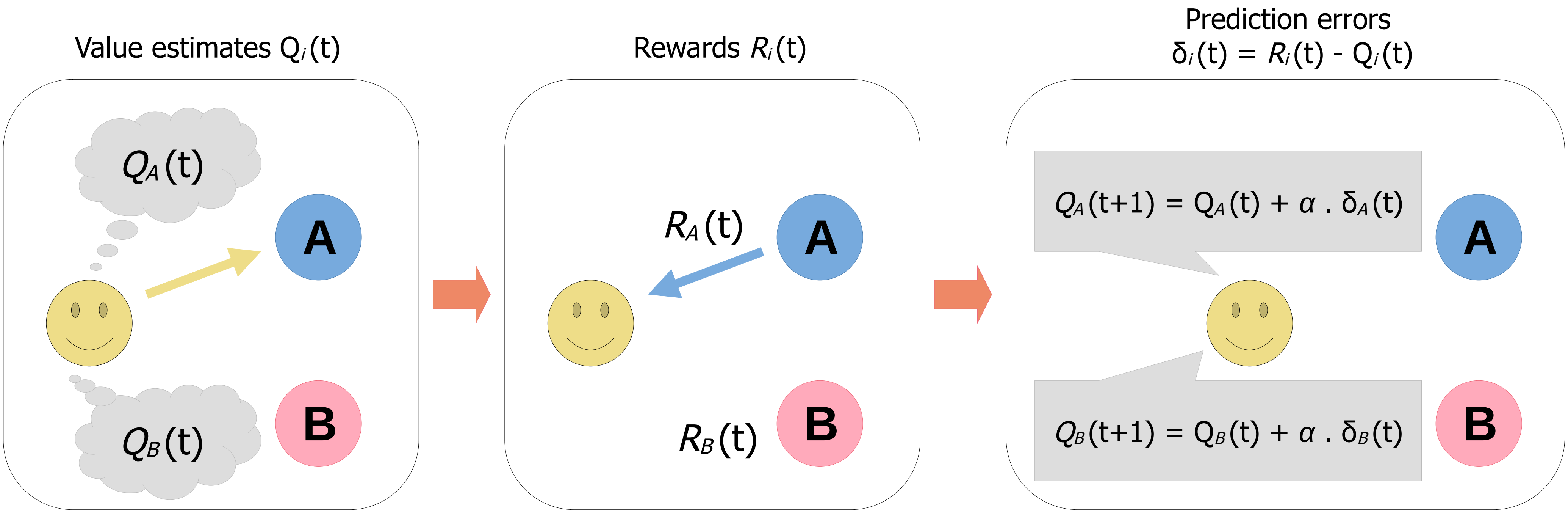


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disconfirmatory prediction errors:

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- $\delta_i(t) \geq 0$ and *i* was unchosen